Philosophy of Science
Prof. James W. McAllister

Lecture 7
Analogies and Models

Motivation of this lecture

- International studies often involves…
  - comparing cases and noting similarities between them
  - creating partial, idealised representations of reality
- Analogies and models are essential conceptual tools for this work
  - Today's lecture is a primer on their use

Plan of this lecture

- Preview of final exam
- What is analogy?
  - Four components: source domain, target domain, mapping, relations
  - Heuristics of analogy: neutral analogy
- What is a model?
  - Models, idealisation, and analogy

Reading for this lecture

- Allan Gibbard and Hal R. Varian, "Economic Models"
  - *Journal of Philosophy* 75 (1978), pp. 664–677
- Available on Brightspace

Preview of final exam

- Two-hour written exam
  - Part 1: twenty multiple-choice questions
  - Part 2: two open essay questions—max. 300 words per question
  - Parts 1 and 2 have equal weight
  - Closed book exam
  - Platform Ans on university Chromebooks
  - See MyTimetable for date, time, place
- We now review some practice questions
  - Three multiple-choice, two open essay

Multiple-choice question 1

Which of the following best characterises the nomothetic approach?

A. It is a model of explanation proposed by Carl G. Hempel in the 1960s
B. It is typical of the natural sciences
C. It cannot be applied to human and social phenomena
D. It is incompatible with induction

Multiple-choice question 2

The correspondence theory of truth says that a statement *P* is true if and only if…

A. We are justified in believing that *P*
B. We know *P* to be true
C. *P* corresponds with the facts
D. We have convincing empirical evidence for *P*

Multiple-choice question 3

What does it mean to say that induction is ampliative?

A. Induction rests on ample empirical evidence
B. Inductive inferences give researchers secure grounds for their conclusions
C. Induction goes from the general to the particular
D. Inductive inferences go beyond their premises

Answers and references

- Question 1: B
  - Lecture 1, sheet 28
- Question 2: C
  - Lecture 2, sheet 20
- Question 3: D
  - Lecture 3, sheet 23

Open essay question 1

A. According to the JTB account of knowledge, under which conditions may we assert "Andrea knows that Bob was born in Paris"? Explain the meaning of any philosophical terms that you use. (5 marks)
B. Assume that Andrea knows that Bob was born in Paris. Then, explain how it is possible for Chris to believe that Bob was not born in Paris, but not possible for Chris to know that Bob was not born in Paris. (3 marks)
C. Is it possible to have a justified belief that *P* where *P* is a proposition that is not true? Explain your reasoning. (2 marks)

Open essay question 2

A. What are the requirements for a successful analogy in science? Explain the meaning of any technical terms that you use. (5 marks)
B. Using the framework that you have just laid out, develop an analogy between an electronic computer and a human brain. (3 marks)
C. How can analogy lead to new discoveries in science? Give an example as part of your answer. (2 marks)

Components of good essays

- Introduction to the philosophical context
- Exposition and explanation of relevant philosophical theories
- Reasoning and arguments establishing pertinent conclusions
- Examples and illustrations
- Discussion of further implications

What is analogy?

- Analogy is…
  - Relation of similarity or isomorphism between two domains of phenomena
- Analogy makes it possible to…
  - Transfer cognitive resources from one domain of science to another
- Analogy helps researchers…
  - Discover similarities and identities between domains
  - Generalise over and unify distinct domains
  - Create and extend scientific concepts

Mary B. Hesse, 1924–2016

- Professor of Philosophy of Science, University of Cambridge
- Author of *Models and Analogies in Science* (Notre Dame University Press, 1966)
- Developed the concepts of positive, negative, and neutral analogy

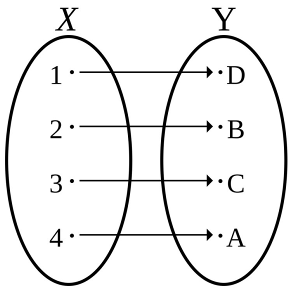
Structure of analogy 1

- What is required for analogy?
  - Four conceptual elements: source domain, target domain, mapping, relations
- Source domain
  - Domain of phenomena that supplies the terms of the analogy: typically more familiar to us
- Target domain
  - Domain of phenomena to which we apply the analogy: typically less familiar to us

Structure of analogy 2

- Mapping
  - Set of correspondences between terms in the source and terms in the target domain
- One or more relations
  - Claims that, under the mapping, hold for both the source and the target domain
- We now put these four components together

Mapping between domains

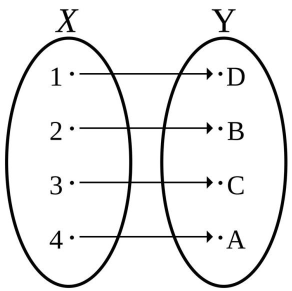


- The mapping sets up a one-to-one correspondence…
- from terms 1–4 of source domain *X*
- to terms *A*–*D* of target domain *Y*

Role of relations

- But a mapping on its own is not enough
  - Merely a translation between terms
  - Says nothing definite about either domain
- We also need one or more relations
  - Causal or structural claims that remain invariant under the mapping
- Relations hold both…
  - between terms in the source domain…
  - and between the corresponding terms in the target domain

Adding relations to the picture



- Causal or structural claims
- E.g. "1, 2, and 3 are causes of 4"
- which, under the mapping, translates to…
- "*D*, *B*, and *C* are causes of *A*"

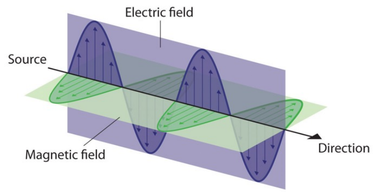
Summarising…

- Analogy requires four components
  - Source and target domains
  - Mapping, which says what in the target domain corresponds to what in the source domain
  - Relations, which say which structures are the same in the two domains
- Two historical examples follow
  - Physics: water and light waves
  - Sociology: organisms and societies

Example 1: physics

- Analogy between waves on the surface of water and light phenomena
- Mapping:
  - Transverse displacement of water molecules → Electric and magnetic field intensity (y)
  - Distance → Distance (x)
  - Time → Time (t)
  - Velocity → Velocity (v)
- Relation:
  - Wave equation: $\partial^2 y/\partial x^2 = (\partial^2 y/\partial t^2)/v^2$

Wave analogy in pictures



- Wave equation describes both water waves and light waves
- It picks out structures common to both

Example 2: sociology

- Early social science understood societies by analogy with organisms
  - Herbert Spencer (1820–1903), British evolutionary theorist
  - Emile Durkheim (1858–1917), French sociologist
  - Functionalism: interpreted society in terms of the functions of its constituent elements
- “Society is analogous to an organism”
  - How to make this analogy more precise? By specifying a mapping and relations

Organic analogy: mapping

- Mapping from domain of organisms to that of societies

  Living organism → Society
  Parts, e.g. organs → Components, e.g. institutions
  Functions of organs → Functions of institutions
  Ill health → Loss of social equilibrium
- The mapping tells us how to translate terms from source to target domains
  - But still it leaves the analogy contentless
  - We have made no claims yet about society

Organic analogies: relations

- Claims that hold in both domains:
  - The whole [organism / society] is greater than the sum of its parts [organs / institutions]
  - Parts [organs / institutions] work together for the benefit of the whole [organism / society]
  - To understand a part [organ / institution] is to show how it relates to other parts and what role it plays in the continued existence of the whole [organism / society]

Heuristics of analogy 1

- Hesse: relations fall into three classes
- Positive analogy
  - Relations that we already know hold in both source and target domains
- Negative analogy
  - Relations that we already know hold in one domain but not the other
- Neutral analogy
  - Relations of which we know that they hold in the source domain, but not whether they hold in the target domain

Heuristics of analogy 2

- Analogical reasoning involves sorting…
  - the neutral analogy into positive or negative
  - That is, exploring whether further relations in the source domain hold also in the target domain
- Neutral analogy ensures the creativity of analogies
  - It generates novel predictions
  - Analogy as quasi-inductive reasoning

Neutral analogy: examples

- Neutral analogy in wave analogies
  - Water waves propagate in a material medium—what about light waves?
  - Research question in 19th-century physics
  - Since then, physicists have moved “material medium” from neutral to negative analogy
- Neutral analogy in organic analogies
  - All organisms age and die—is societal collapse inevitable?
  - New research question for sociology

Negative analogy: significance

- Discovery of a negative analogy…
  - can also be a progressive development, as the physics example showed
- Existence of negative analogy reinforces the intuition that analogies…
  - break down beyond some point
  - are pragmatically useful or illuminating, rather than true or false
- Emphasises distinctiveness of the relata

What is a model?

- A scientific model is…
  - A simplified representation of a domain of phenomena
- Models make possible…
  - The application of a scientific theory to a complex phenomenon
- Link with analogy
  - Many models are based on analogies

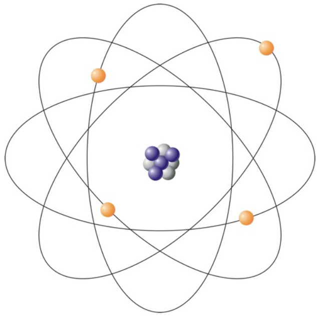
Models and idealisation

- Models are idealisations of phenomena
  - They deliberately simplify and distort reality
  - Aim: to make phenomena more tractable
  - Examples: frictionless planes, perfectly rational agents
- Researchers normally take an instrumentalist stance towards models
  - They regard models as adequate or inadequate, not as true or false
- Two examples follow

Models: example 1

- Bohr model of the atom
  - Formulated in 1913
  - Niels Bohr (1885–1962), Danish physicist
- Depicts the atom as composed of:
  - small, positively charged nucleus
  - electrons in orbit around the nucleus
- Inspired by an analogy with the solar system in astronomy

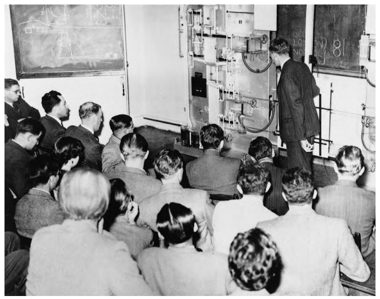
Bohr model of the atom



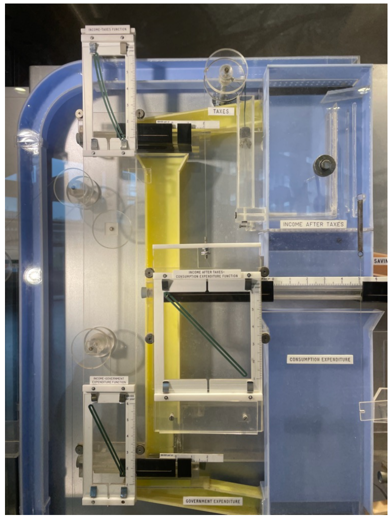
- Small nucleus made of protons and neutrons at centre
- Electrons in orbit around the nucleus
- The Bohr model has now been replaced by quantum models, in which electron trajectories are not localised

Models: example 2

- Monetary National Income Analogue Computer or MONIAC
  - Hydraulic simulation of the UK economy
  - Implements tenets of Keynesian economics
  - Built by Bill Phillips in 1949 while a student at the London School of Economics (LSE)
- Structure of tanks, pipes, drains, pumps
  - Water flows round the machine
  - Between reservoirs "treasury", "education", "healthcare", "private investment", etc.



MONIAC in an LSE economics seminar, 1950s, left; on display at the Erasmus Universiteit Rotterdam, 2020, right



MONIAC, reservoirs representing taxes

MONIAC and analogy

- MONIAC embodies an analogy between the hydraulic system and the UK economy
- Mapping:
  - Water → Money
  - Tanks → Economic sectors
  - Pipes → Financial relationships
  - Pumps → Mechanisms for transferring funds
- Relations:
  - A particular configuration [flow pattern / economic regime] is stable

Recapitulation

- Analogies
  - What is analogy?
  - Four components: source domain, target domain, mapping, relations
  - Heuristics of analogy: neutral analogy
- Models
  - What is a model?
  - Models and analogy
  - Models and idealisation